

HS 312 – Introduction to Science and Technology Studies

Lecture 5

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Outline

- What makes knowledge scientific?
- Science = Progress?

Rethinking history of science

- Thomas Kuhn – *The Structure of Scientific Revolutions* (1962)
- Science is merely what scientist do
- Too simple **progressivism** – attempts to construct/see what we now believe to be the truth as more rational, more natural, teleological
- Normal Science – is the science done when members of a field share a recognition of beliefs about theories, important problems, methods for solving

Normal Science

- Classical or Newtonian physics paradigm – preference for causal explanations, precise quantitative predictions, world as composed of material particles (that collide due to forces acting in straight line), laws of motion and gravitation, standard mathematical techniques, accounts for friction/resistance
- Paradigm examples – Ptolemaic astronomy, **Phlogiston theory of combustion**, Daltonian chemistry (theory of differing atomic weight of elements), fluid flow theory of electricity (electricity is a material fluid), caloric theory of heat (heat is a material fluid), particle optics (light is a collective of fast moving particles), relativistic physics (time between events is relative to frame of reference), quantum physics (energy possessed by objects/waves comes in discrete units)

Normal Science

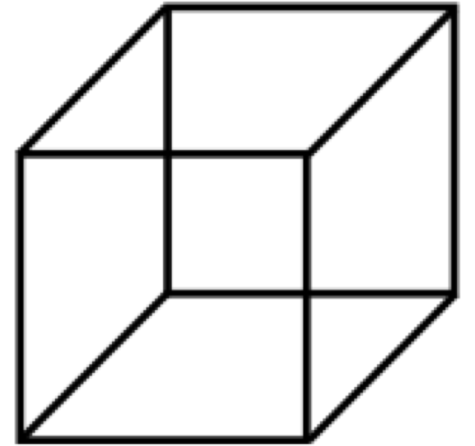
- **Puzzle-solving** – everyday practice of science is a conservative activity, problems are to be solved within the terms/rules of the paradigm; unsolved problem is an anomaly
- In periods of normal science the paradigm is not open to serious question
- Socializing practitioners – Students learn from textbooks; they have lengthy periods of training, and are generally asked to solve well-understood and well-structured problems
- We can talk of progress as worldviews and frameworks are shared

Revolution

- Period of *crisis* – anomalies accumulate and become real problems; scientists question the fundamental principles of their discipline, and might consider alternatives to the framework;
- If alternative is created, young scientist adopt it
- Non-progressive – **revolutions build and destroy**, as pre-revolutionary paradigm will fail to make sense in new
- **Incommensurable** – the meanings of theoretical terms change with revolutions

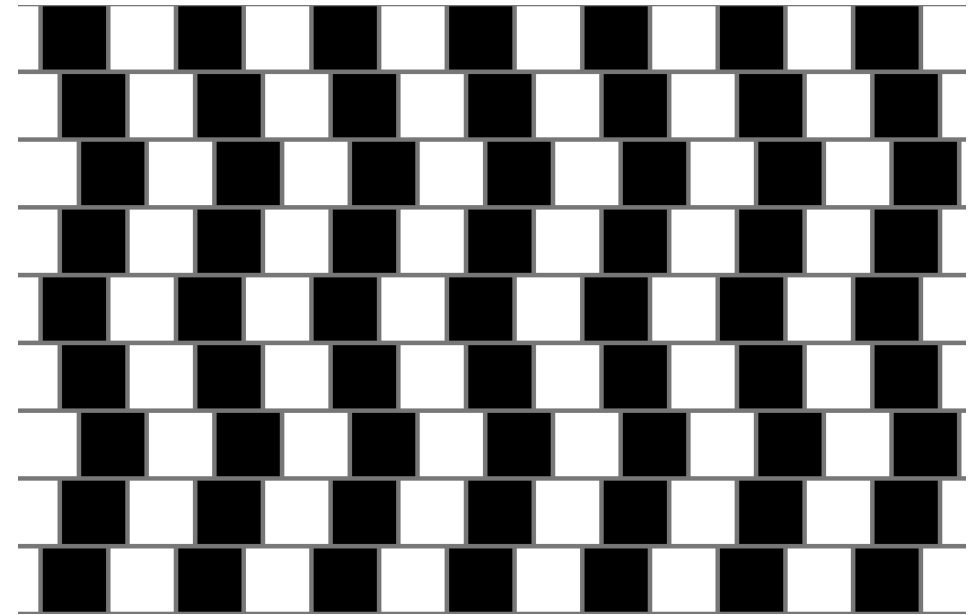
Revolution

- SSR showed that science primarily advances through shifts in theory and not by the accumulation of knowledge
i.e. shift from one adequate paradigm to another
- **Radical implication** – Science creates different partial views that can be considered to contain truth only by people who hold those views!



Revolution

- Paradigms change should result in widespread changes in the meanings of key terms; none from one paradigm would map neatly onto those of another
- Indoctrination – paradigms even shape observations; no raw observation - we do not see dots and lines in our visual fields, but instead see more or less recognizable objects
- **Theory-dependence of observation** – during revolutions people stop seeing one way, and start seeing another way (like a Gestalt)



Objections to Kuhn's thesis

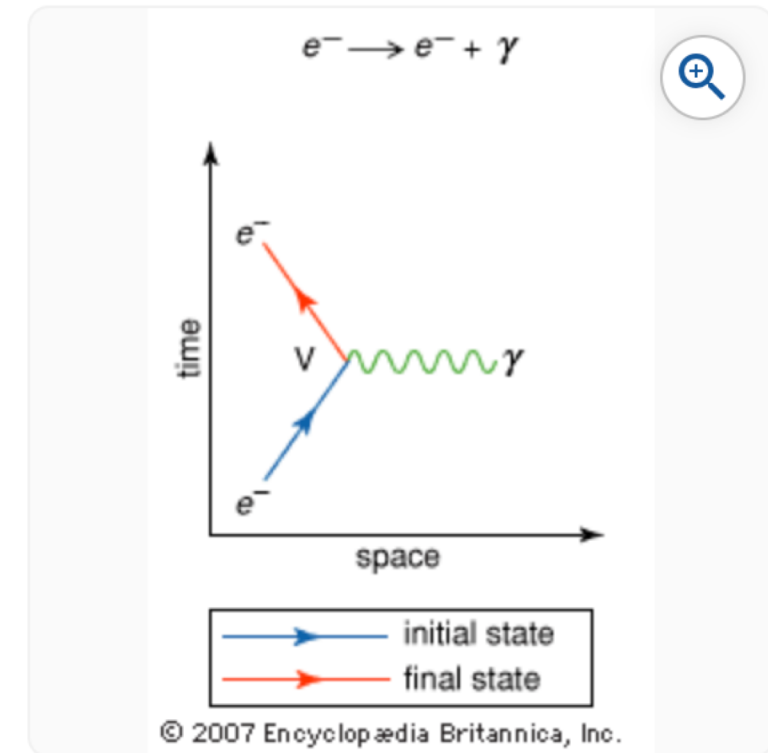
- Semantic incommensurability – Meanings of terms do change, but not so much that claims in which they are used cannot typically be compared
- Kuhn – incommensurability as ‘incomplete communication’ or ‘difficulty of translation’
- If no radical incommensurability then no radical paradigm division
- Key components rarely change together – instruments, theories, and experiments change at different times
- Science thus maintains stability and *ad hoc* unity

Communicating Among Social Worlds

- To what extent people across disciplines/boundaries can communicate?
- **Epistemic cultures** – different orientation to object of research, patterns of interaction, and meanings e.g. molecular biology vs. evolutionary ecology vs. neuropathology
- The question is – How people from different areas interact, and produce a degree of unity for science?

Communicating Among Social Worlds

- *Trading zones* – an area where scientific practices interact via simplified languages or *pidgins* e.g. Feynman diagrams
- Specialities can collaborate even if cultures and practices do not agree on problems or definitions
- *Boundary objects* – objects can form bridges across boundaries, allow for a certain amount of coordination of actions without large measures of translation e.g. sketches and engg drawings
- Incommensurability sometimes thus can be overcome
- Though sciences are disunified but has some stability – the question is how?



Feynman diagram Feynman diagram of the interaction of an electron with the electromagnetic forceThe basic vertex...(more)

Summary

- SSR: Scientific communities are importantly organized around ideas and practices, not around ideals of behaviour
- Organised from the bottom up (not as functionalism argues)
- Changes in theories driven by changes in vision (and not data, as positivism argues)
- Anomalies are typically set aside (not as falsification argues) except during revolutions
- History of science is not an uninterrupted story of progress but of only change
- Thomas Kuhn's SSR has profoundly affected how we think about and study science and technology

Thank you